Ranks of Cohen-Macaulay Modules

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Given a polynomial ring $R = k[x_0, x_1, x_2, \ldots, x_n]$ and a homogeneous polynomial $f \in R$, presence of non-free maximal Cohen-Macaulay modules (MCMs) over $R/(f)$ indicates existence of singularity, by Auslander-Buchsbaum formula. Buchweitz-Greuel-Schreyer, in [5], conjectured that for a regular local ring R of dimension $n+1$ such that $R/(f)$ has an isolated singularity, the rank of any non-free MCMs over $R/(f)$ is at least $2^{\lfloor \frac{n}{2} \rfloor}$. This implies a weaker conjecture by Brown and Walker (see conjecture 1.1. in [4]) that given any nontrivial matrix factorization of f consisting of finite rank free R modules M and M' together with maps $d : M \to M'$ and $d' : M' \to M$ such that $dd' = f \cdot 1_{M'}$ and $d'd = f \cdot 1_M$, the rank of module M is at least $2^{\lfloor \frac{n}{2} \rfloor}$. The connection, suggested by the two conjecture, is made clear by the equivalence (as triangulated categories) of the homotopy category of $\mathbb{Z}/2\mathbb{Z}$-graded matrix factorizations $\mathrm{HMF}_{gr}(f)$ and the stable category of $\mathbb{Z}/2\mathbb{Z}$ MCMs over $R/(f)$. As of now, the conjectures have not been proven independently.

Earlier, in [6], Dyckerhoff showed that starting with local regular ring R with residue field k and maximal ideal $\mathfrak{m}$, given an element $f \in R$, the category $\mathrm{MF}^{\infty}(f)$ of matrix factorizations of f of potentially infinite rank is compactly generated by a stabilization k^{stab} of the residue field. The result of our interest is the quasiequivalence of $\mathrm{MF}^{\infty}(f)$ with an explicit model of the derived category of $\mathrm{End}(k^{stab})^{op}$-modules. This is a new setting in which to investigate the BGS and Brown-Walker conjecture. Our objective is to translate and investigate the conjectures in this setting.

A more modest objective is that there is an A_∞ algebra structure on the algebra $A = \mathrm{End}(k^{stab})^{op}$, which as explained in [6], allows one to see that $MF^{\infty}(f)$ is quasiequivalent to the derived category of modules over $H^*(A)$. However, the A_∞ structure is tedious to compute in practice, as in [6]. We will try to extend the known computations from the case of quadratic forms in $k[x_1, x_2, \ldots, x_n]$ where $\mathrm{char}(k) \neq 2$ to hypersurfaces in general.

Suggested one-page rewrite: Rank Conjectures through Local and Graded A_∞-Models

Background and objective. Let k be an algebraically closed field of characteristic zero, $P = k[x_0, \dots, x_n]$, and $f \in P$ a homogeneous polynomial of degree $d \geq 2$ such that $X = V_+(f) \subset \mathbb{P}^n$ is a smooth projective hypersurface. Its affine cone has an isolated singularity at the vertex $\mathfrak{m}$. Set $Q = \widehat{P_{\mathfrak{m}}}$, $S = Q/(f)$, and $e = \lfloor n/2 \rfloor$. The Buchweitz–Greuel–Schreyer conjecture predicts that every nonfree maximal Cohen–Macaulay (MCM) S-module has rank at least 2^{e-1} [5]; its Brown–Walker weakening predicts that every nontrivial matrix factorization $F = (F^0 \rightleftarrows F^1)$ over Q satisfies $\operatorname{rank}_Q F^0 \geq 2^e$ [4].
There is also a genuinely graded, projective problem. A rank-r Ulrich bundle $\mathcal{E}$ on X corresponds to a graded MCM $P/(f)$-module with a linear P-free presentation

$$0 \longrightarrow P(-1)^{rd} \xrightarrow{\phi} P^{rd} \longrightarrow M \longrightarrow 0;$$

ϕ and a degree-$(d-1)$ companion form a graded matrix factorization of f [2]. Thus its matrix size is rd, whereas the bundle rank is r. The Ulrich complexity $\operatorname{uc}(X)$ is the least such r, and the codimension conjecture specializes in this smooth case to $\operatorname{uc}(X) \geq 2^{\lceil (n+1)/2 \rceil - 2} = 2^{e-1}$ [3]. My objective is to formulate these three lower bounds in suitable A_∞ settings while retaining the distinct rank information that each one measures.
Two A_∞ settings. For the local conjectures, Dyckerhoff showed that $K = k^{\mathrm{stab}}$ compactly generates $\mathrm{MF}^\infty(f)$ over Q [6]. The dg algebra $E_{\mathrm{loc}} = \operatorname{End}(K)^{\mathrm{op}}$ therefore gives a Morita model, and transfer to $H^*(E_{\mathrm{loc}})$ gives a minimal A_∞-algebra. After replacing a finite-rank factorization F by its minimal representative, it determines a perfect A_∞-module P_F with

$$\dim_k H^*(P_F) = 2 \operatorname{rank}_Q(F^0),$$

so the Brown–Walker conjecture becomes $\dim_k H^*(P_F) \geq 2^{e+1}$. Recovering the rank of one MCM cokernel requires finer information than this total dimension.
Dyckerhoff's local, two-periodic model does not retain the internal grading needed for Ulrich bundles. I will instead use the $\mathbb{G}_m$-equivariant, or $\mathbb{Z}$-graded, matrix-factorization category of f. For a smooth projective hypersurface, Ballard–Deliu–Favero–Isik–Katzarkov construct a graded local generator and transfer its endomorphism dg algebra to an explicit, internally graded minimal A_∞-algebra whose degree-d higher product records f [1]. In this fixed-hypersurface setting, perfect internally graded modules over this algebra model graded factorizations and supply the appropriate setting for the Ulrich bound.
Comparison, research questions, and method. The two models are linked by forgetting the internal grading of a graded factorization and completing at the cone vertex. I will transport this functor through the two Morita descriptions. If $\mathcal{E}$ is an Ulrich bundle of rank r, its minimal graded factorization has size rd, so its underlying local perfect module satisfies

$$\dim_k H^*(P_{\mathcal{E}}^{\mathrm{loc}}) = 2rd.$$

Thus the Ulrich lower-bound conjecture becomes $\dim_k H^*(P_{\mathcal{E}}^{\mathrm{loc}}) \geq d2^e$ within the class admitting a linear graded lift, whereas Brown–Walker requires $\dim_k H^*(P_F) \geq 2^{e+1}$ for every nontrivial minimal local factorization. Total dimension detects matrix size, but recovering r requires the degree and linear graded presentation; recovering the rank of one MCM cokernel requires still finer, parity-sensitive information.
The categorical equivalences do not automatically preserve these numerical features: stabilization suppresses free summands, while completion forgets internal degrees. I will therefore ask: (i) what data on a minimal perfect A_∞-module distinguish the ranks of its two MCM cokernels; (ii) what bigraded generation or vanishing condition characterizes the linear presentation of an Ulrich bundle; and (iii) whether the higher module operations obstruct objects below the conjectured bounds. Since the degree-d multiplication records f, the A_∞-module identities turn the last question into a concrete obstruction problem.
For each test case, I will start with an explicit minimal factorization, apply the relevant generator functor, and compute its transferred module structure. I will compare parity, total cohomology, internal Hilbert series, and minimal generator degrees with the original matrix size, MCM rank, and Betti numbers, and verify their behaviour under quasi-isomorphism and grading shifts. Quadrics provide a consistency check through Clifford algebras and spinor bundles; one or two low-dimensional smooth cubic or quartic examples with known factorizations will supply the main computations. The deliverables will be a precise dictionary for the three rank questions, a proposed graded criterion for recognizing Ulrich objects, and worked examples identifying which rank and linearity information survives each categorical passage.

References

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